1 Methodology

We study continual learning for autoregressive neural surrogates of one-dimensional partial differential equations (PDEs). The model is a Fourier Neural Operator (FNO1D) that receives a short temporal context and predicts the next field state. During training, the model is exposed sequentially to experiences corresponding to different physical regimes. After each experience, it is evaluated on the full stream, allowing us to measure both adaptation to the current regime and retention of previously observed regimes.

We compare six continual-learning strategies: Naive sequential fine-tuning, uniform Experience Replay, Replay-TS, Gradient Episodic Memory (GEM), Averaged GEM (A-GEM), and Elastic Weight Consolidation (EWC). Naive fine-tuning updates only on the current experience. Replay methods retain a bounded memory of previous samples; Replay-TS additionally selects temporal target slices. GEM and A-GEM constrain updates using gradients from memory, while EWC regularizes changes to parameters important to earlier experiences.

2 Experimental Setup

The experiments use a mixed stream of AdvDiff1D and Burgers1D data. The stream contains four alternating experiences: AdvDiff1D, Burgers1D, AdvDiff1D, and Burgers1D. This ordering tests whether a method can alternate between distinct PDE families without catastrophically forgetting the other family. All methods use the same FNO1D architecture, random initialization, optimizer, seed, and CPU device.

Because the full benchmark is expensive, we first run a reproducible test. Each simulation contributes 16 snapshots, the autoregressive context length is 3, and the rollout length is 2. Training uses one epoch per experience and batch size 32. These settings are intended for method screening and should not be interpreted as final benchmark settings.

We report the mean stream MSE and mean relative L^2 error after the final experience. For a stream with experiences $e = 1, \ldots, E$, the reported quantities are

$$\mathrm{MSE}_{\mathrm{stream}} = \frac{1}{E}\sum_{e=1}^{E} \mathrm{MSE}_e, \qquad \mathrm{RelL2}_{\mathrm{stream}} = \frac{1}{E}\sum_{e=1}^{E} \left(\frac{\|\hat{u}_e - u_e\|_2^2}{\|u_e\|_2^2} \right)^{1/2}. \quad (1)$$

3 Experimental Results

Table 1 summarizes the final mixed-stream test results. Lower values are better for both metrics. Replay-TS obtains the best relative L^2 error and the best MSE, closely followed by A-GEM and Replay. The three methods that explicitly use past-experience information outperform Naive, GEM, and EWC in this reduced experiment. This result is suggestive rather than conclusive because it uses one seed, one epoch, and a shortened rollout.

Table 1: Final results on the mixed AdvDiff1D–Burgers1D CPU test.

Method	Stream MSE	Relative L^2
Naive	0.004288	0.07573
Replay	0.000769	0.03302
Replay-TS	0.000595	0.02910
GEM	0.003551	0.06975
A-GEM	0.000650	0.03015
EWC	0.003861	0.07194

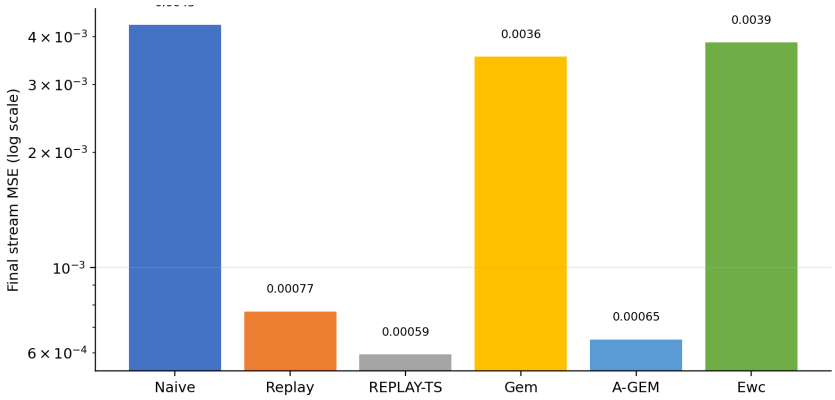


Figure 1: Final stream MSE for the mixed AdvDiff1D-Burgers1D test.

The table shows that Replay-TS is the strongest method in this initial comparison, reducing the final relative L^2 error by approximately 62% relative to Naive fine-tuning. The next step is to repeat the comparison across multiple seeds and increase the number of epochs, snapshots, and rollout steps before drawing conclusions about the full benchmark.

The corresponding error plots are shown in Figures 1 and 2.

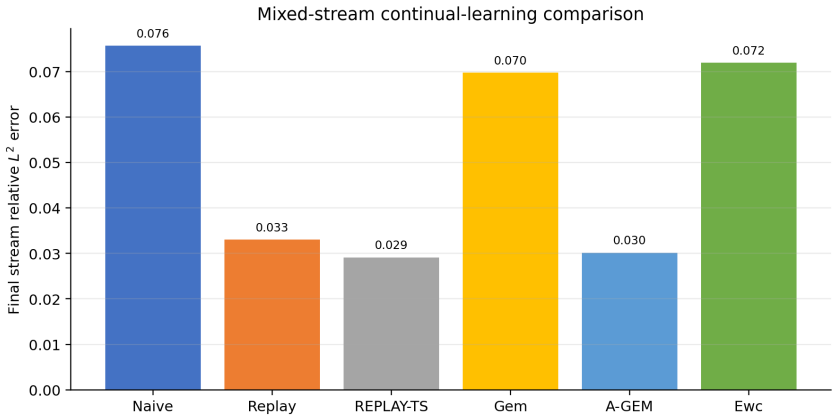


Figure 2: Final stream relative L^2 error for the mixed AdvDiff1D–Burgers1D test.